

MCDONALD'S ICE CREAM MACHINES

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INTRODUCTION

McDonald's ice cream machines are known for their constant breakdowns, we want to know if machines could be better in Wisconsin.

To do this, we want to compare the proportion of broken ice cream machines in Wisconsin to other states and see if it is lower in Wisconsin.

In this project, we will answer this question by conducting a two sample z test to see if there is evidence to support our claim



ORIGINAL DATA

- Data was taken from Kaggle (theDevastator and McBroken website) and is a csv file of every ice cream machine in the country, location, status, coordinates and many other variables.

lat	lon	alt	is_broken	is_active	dot	state	city	street	country	last_checked
-73.988281	40.71883	0	FALSE	TRUE	working	NY	New York	114 Delancey St	USA	Checked 142 minutes ago
-74.00509	40.728794	0	FALSE	TRUE	working	NY	New York	208 Varick St	USA	Checked 142 minutes ago
-73.993408	40.729197	0	FALSE	TRUE	working	NY	New York	724 Broadway	USA	Checked 142 minutes ago
-73.985855	40.726555	0	FALSE	TRUE	working	NY	New York	102 1st Ave	USA	Checked 142 minutes ago
-73.991692	40.691383	0	TRUE	TRUE	broken	NY	Brooklyn	82 Court St	USA	Checked 142 minutes ago
-73.982277	40.731014	0	FALSE	TRUE	working	NY	New York	404 E 14th St	USA	Checked 142 minutes ago
-73.996726	40.737864	0	FALSE	TRUE	working	NY	New York	541 6th Ave	USA	Checked 142 minutes ago
-73.990517	40.736977	0	TRUE	TRUE	broken	NY	New York	39 Union Square W	USA	Checked 142 minutes ago

CLEANED DATA

We cleaned the data to hold only two variables:

- Whether the machine is broken
- State in which it is located

We also eliminated machines that are not in the United States and machines that are inactive all together.

	is_broken	state
1	FALSE	NY
2	FALSE	NY
3	FALSE	NY
4	FALSE	NY
5	TRUE	NY
6	FALSE	NY
7	FALSE	NY
8	TRUE	NY
9	FALSE	NJ
10	TRUE	NJ
11	TRUE	NY
12	FALSE	NY
13	TRUE	NY
14	FALSE	NY
15	TRUE	NJ
16	TRUE	NJ

VISUALIZING OUR PROBLEM

Proportion of Broken Ice Cream machines in Wisconsin v The Other 49 states

Extracted on November, 24, 2024 at 19:37

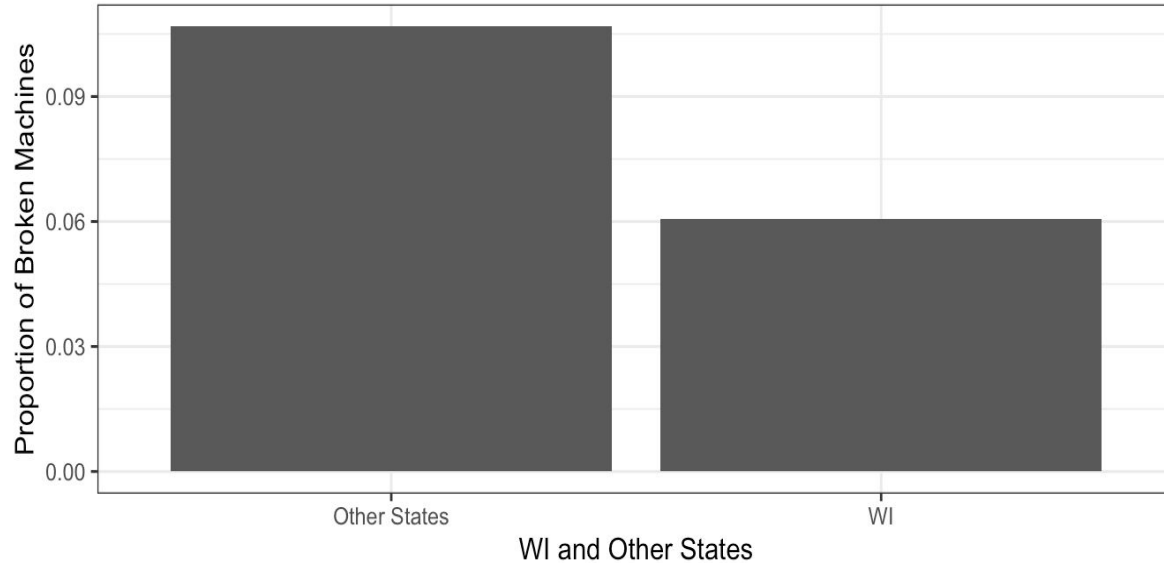


Figure 1: Bar Chart of Proportions of Broken Machines in dataset

ANALYSIS AND RESULTS

- Our analysis is a two sided z test checking the difference in proportions of broken machines in Wisconsin and outside Wisconsin
- Our analysis yielded a p-value of .00983 which was lower than our chosen alpha of 0.05.
- Therefore, We have evidence to suggest that the proportion of broken machines in WI is lower than other states.

LIMITATIONS AND FUTURE SCOPE

- Our data is based on customer complaints and can be skewed by areas with extremely low or high number of complaints
 - The data is from a snapshot in time and doesn't account for the behavior of machines outside of that.
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- Why is the proportion of broken machines in WI lower?
 - Does it have something to do with WI being the “dairy state”?
 - Further analysis could explain deeper trends in the data.

